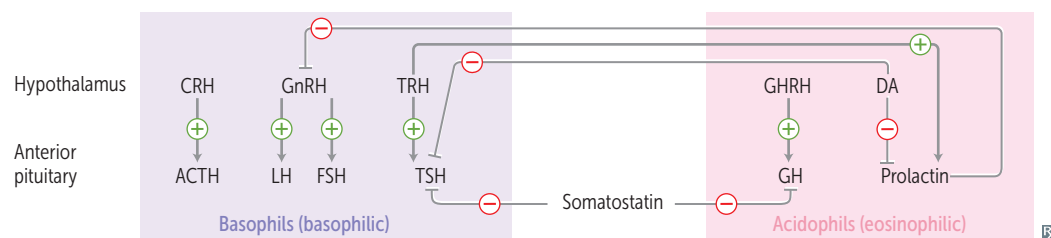


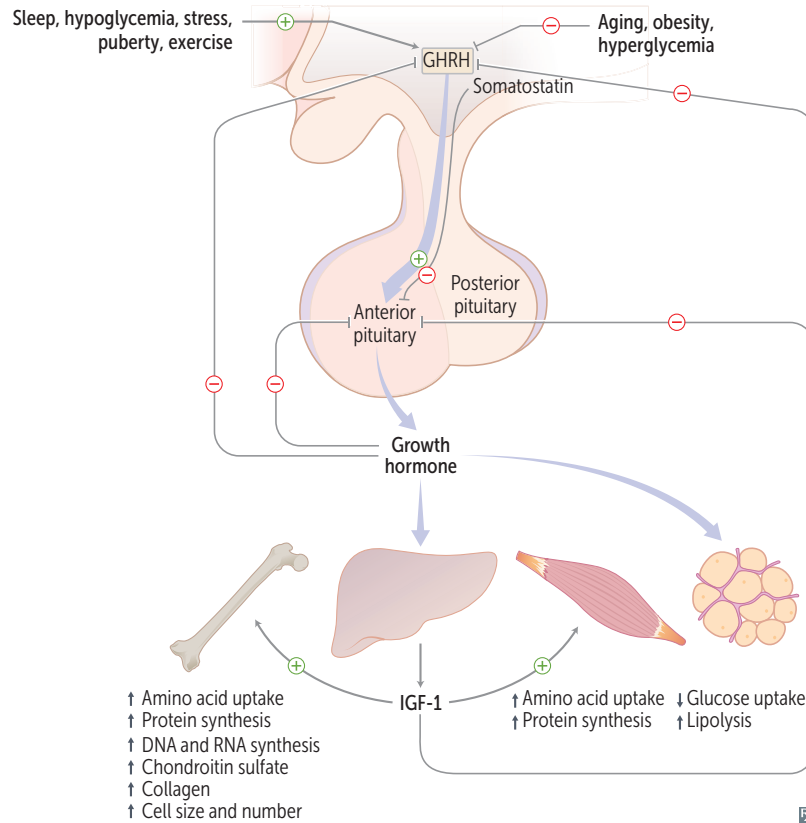
► ENDOCRINE—PHYSIOLOGY

Hypothalamic-pituitary hormones

HORMONE	FUNCTION	CLINICAL NOTES
ADH	↑ water permeability of distal convoluted tubule and collecting duct cells in kidney to ↑ water reabsorption	Alcohol consumption → ↓ ADH secretion → polyuria and dehydration
CRH	↑ ACTH, ↑ MSH, ↑ β-endorphin	↓ in chronic glucocorticoid use
Prolactin-inhibiting factor	↓ prolactin, ↓ TSH	Commonly referred to as dopamine Dopamine antagonists (eg, antipsychotics) can cause galactorrhea due to hyperprolactinemia
GHRH	↑ GH	Analogue (tesamorelin) used to treat HIV-associated lipodystrophy
GnRH	↑ FSH, ↑ LH	Suppressed by hyperprolactinemia Tonic GnRH analogue (eg, leuprolide) suppresses hypothalamic–pituitary–gonadal axis. Pulsatile GnRH leads to puberty, fertility
MSH	↑ melanogenesis by melanocytes	Causes hyperpigmentation in Cushing disease, as MSH and ACTH share the same precursor molecule, POMC
Oxytocin	Causes uterine contractions during labor. Responsible for milk letdown reflex in response to suckling.	Modulates fear, anxiety, social bonding, mood, and depression
Prolactin	↓ GnRH Stimulates lactogenesis.	Pituitary prolactinoma → amenorrhea, osteoporosis, hypogonadism, galactorrhea Breastfeeding → ↑ prolactin → ↓ GnRH → delayed postpartum ovulation (natural contraception)
Somatostatin	↓ GH, ↓ TSH	Also called growth hormone inhibiting hormone (GHIH)
TRH	↑ TSH, ↑ prolactin	↑ TRH (eg, in 1°/2° hypothyroidism) may increase prolactin secretion → galactorrhea



Growth hormone



Also called somatotropin. Secreted by anterior pituitary.

Stimulates linear growth and muscle mass through IGF-1 (somatomedin C) secretion by liver. ↑ insulin resistance (diabetogenic).

Released in pulses in response to growth hormone–releasing hormone (GHRH).

Secretion ↑ during (deep) sleep, hypoglycemia, stress, puberty, exercise.

Secretion ↓ with aging, obesity, hyperglycemia, somatostatin, somatomedin (regulatory molecule secreted by liver in response to GH acting on target tissues).

Excess secretion of GH (eg, pituitary adenoma) may cause acromegaly (adults) or gigantism (children). Treatment: somatostatin analogs (eg, octreotide) or surgery.

Antidiuretic hormone

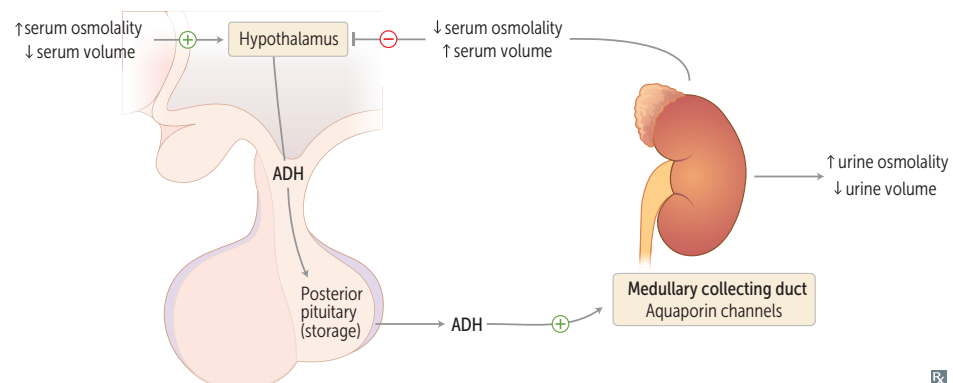
Also called vasopressin.

SOURCE	Synthesized in hypothalamus (supraoptic and paraventricular nuclei), stored and secreted by posterior pituitary.
FUNCTION	Regulates blood pressure (V_1 -receptors) and serum osmolality (V_2 -receptors). Primary function is serum osmolality regulation (ADH ↓ serum osmolality, ↑ urine osmolality) via regulation of aquaporin channel insertion in principal cells of renal collecting duct.
REGULATION	Plasma osmolality (1°); hypovolemia.

ADH level is ↓ in central diabetes insipidus (DI), normal or ↑ in nephrogenic DI.

Desmopressin (ADH analog) is a treatment for central DI and nocturnal enuresis.

Vasopressin is a potent **vasopressor** that can be used to increase organ perfusion in septic shock.



Prolactin

SOURCE

Secreted mainly by anterior pituitary.

Structurally homologous to growth hormone.

FUNCTION

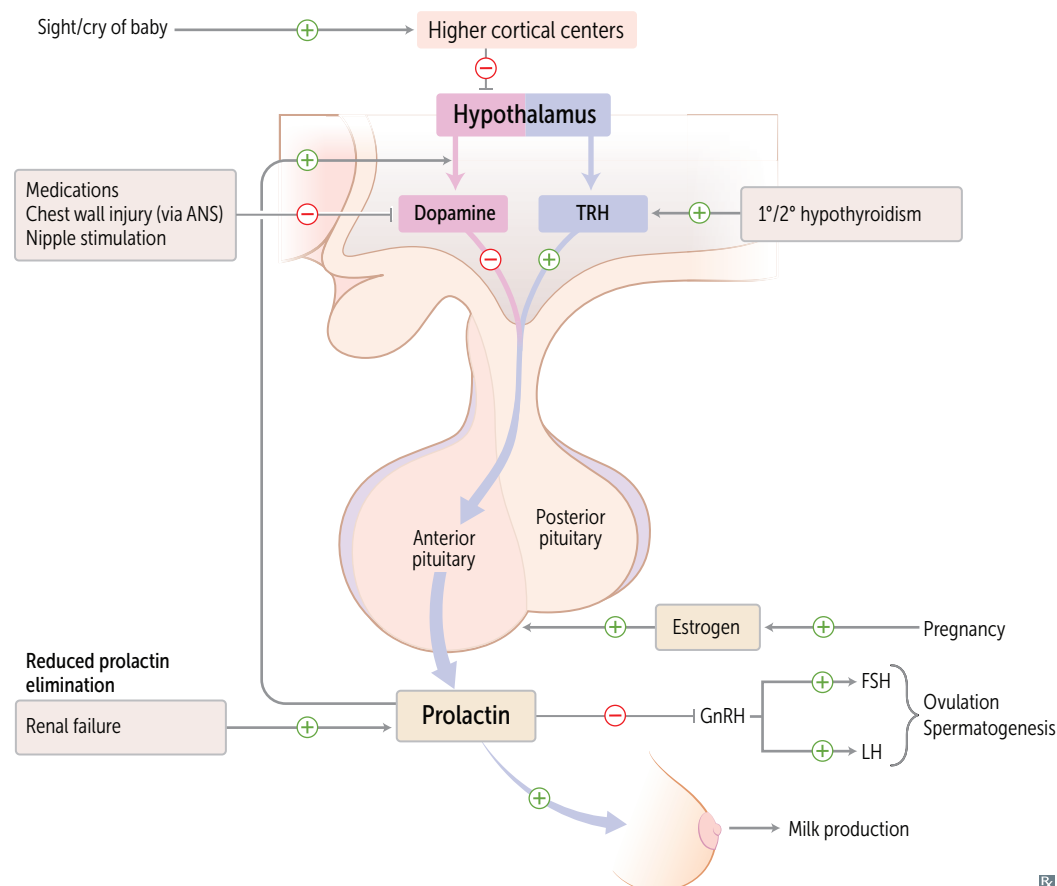
Stimulates milk production in breast; inhibits ovulation in females and spermatogenesis in males by inhibiting GnRH synthesis and release.

Excessive amounts of prolactin associated with galactorrhea, amenorrhea, infertility, osteoporosis (in females); gynecomastia, infertility, decreased libido, impotence (in males).

REGULATION

Prolactin secretion from anterior pituitary is tonically inhibited by dopamine from tuberoinfundibular pathway of hypothalamus. Prolactin in turn inhibits its own secretion by $\uparrow$ dopamine synthesis and secretion from hypothalamus. TRH $\uparrow$ prolactin secretion (eg, in 1° or 2° hypothyroidism). Dopamine has stronger effect on prolactin regulation than TRH does.

Dopamine agonists (eg, bromocriptine, cabergoline) inhibit prolactin secretion and can be used in treatment of prolactinoma. Dopamine antagonists (eg, most antipsychotics, metoclopramide) and estrogens (eg, OCPs, pregnancy) stimulate prolactin secretion.



Thyroid hormones

Thyroid produces triiodothyronine (T_3) and thyroxine (T_4), iodine-containing hormones that control the body's metabolic rate.

SOURCE

Follicles of thyroid. 5'-deiodinase converts T_4 (the major thyroid product) to T_3 in peripheral tissue (5, 4, 3). Peripheral conversion is inhibited by glucocorticoids, β -blockers, and propylthiouracil (PTU). Reverse T_3 (rT_3) is a metabolically inactive byproduct of the peripheral conversion of T_4 and its production is increased by growth hormone and glucocorticoids. Functions of thyroid peroxidase include oxidation, organification of iodine, and coupling of monoiodotyrosine (MIT) and diiodotyrosine (DIT). Inhibited by PTU and methimazole. $DIT + DIT = T_4$. $DIT + MIT = T_3$. Wolff-Chaikoff effect—protective autoregulation; sudden exposure to excess iodine temporarily turns off thyroid peroxidase $\rightarrow$ $\downarrow T_3/T_4$ production.

FUNCTION

Only free hormone is active. T_3 binds nuclear receptor with greater affinity than T_4 . T_3 functions —7 B's:

- **B**rain maturation
- **B**one growth (synergism with GH and IGF-1)
- **β** -adrenergic effects. $\uparrow \beta_1$ receptors in heart $\rightarrow$ $\uparrow$ CO, HR, SV, contractility; β -blockers alleviate adrenergic symptoms in thyrotoxicosis
- **B**asal metabolic rate $\uparrow$ (via $\uparrow Na^+/K^+$ -ATPase $\rightarrow$ $\uparrow O_2$ consumption, RR, body temperature)
- **B**lood sugar ($\uparrow$ glycogenolysis, gluconeogenesis)
- **B**reak down lipids ($\uparrow$ lipolysis)
- Stimulates surfactant synthesis in **B**abies

REGULATION

TRH $\rightarrow$ $\oplus$ TSH release $\rightarrow$ $\oplus$ follicular cells. Thyroid-stimulating immunoglobulin (TSI) may $\oplus$ follicular cells in Graves disease.

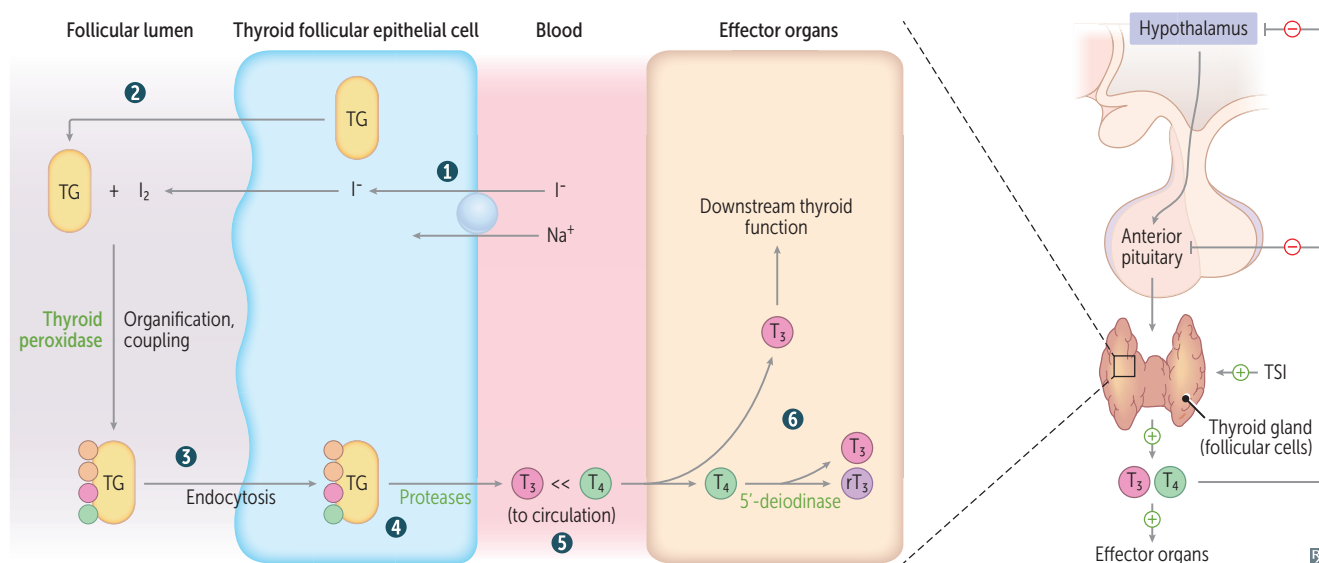
Negative feedback primarily by free T_3/T_4 :

- Anterior pituitary $\rightarrow$ $\downarrow$ sensitivity to TRH
- Hypothalamus $\rightarrow$ $\downarrow$ TRH secretion

Thyroxine-binding globulin (TBG) binds most T_3/T_4 in blood. Bound T_3/T_4 = inactive.

- $\uparrow$ TBG in pregnancy, OCP use (estrogen $\rightarrow$ $\uparrow$ TBG) $\rightarrow$ $\uparrow$ total T_3/T_4
- $\downarrow$ TBG in steroid use, nephrotic syndrome

T_3 and T_4 are the only lipophilic hormones with charged amino acids and require specific transporters to diffuse into the cell (facilitated diffusion).



Parathyroid hormone

SOURCE

Chief cells of parathyroid

FUNCTION

$\uparrow$ free Ca^{2+} in the blood (1° function)
 $\uparrow$ Ca^{2+} and PO_4^{3-} absorption in GI system
 $\uparrow$ Ca^{2+} and PO_4^{3-} from bone resorption
 $\uparrow$ Ca^{2+} reabsorption from DCT
 $\downarrow$ PO_4^{3-} reabsorption in PCT
 $\uparrow$ $1,25\text{-(OH)}_2\text{D}_3$ (calcitriol) production by activating 1α -hydroxylase in **PCT** (tri to make D_3 in the **PCT**)

PTH $\uparrow$ serum Ca^{2+} , $\downarrow$ serum PO_4^{3-} , $\uparrow$ urine PO_4^{3-} , $\uparrow$ urine cAMP
 $\uparrow$ RANK-L (receptor activator of NF- κ B ligand) secreted by osteoblasts and osteocytes; binds RANK (receptor) on osteoclasts and their precursors to stimulate osteoclasts and $\uparrow$ Ca^{2+} $\rightarrow$ bone resorption (intermittent PTH release can also stimulate bone formation)

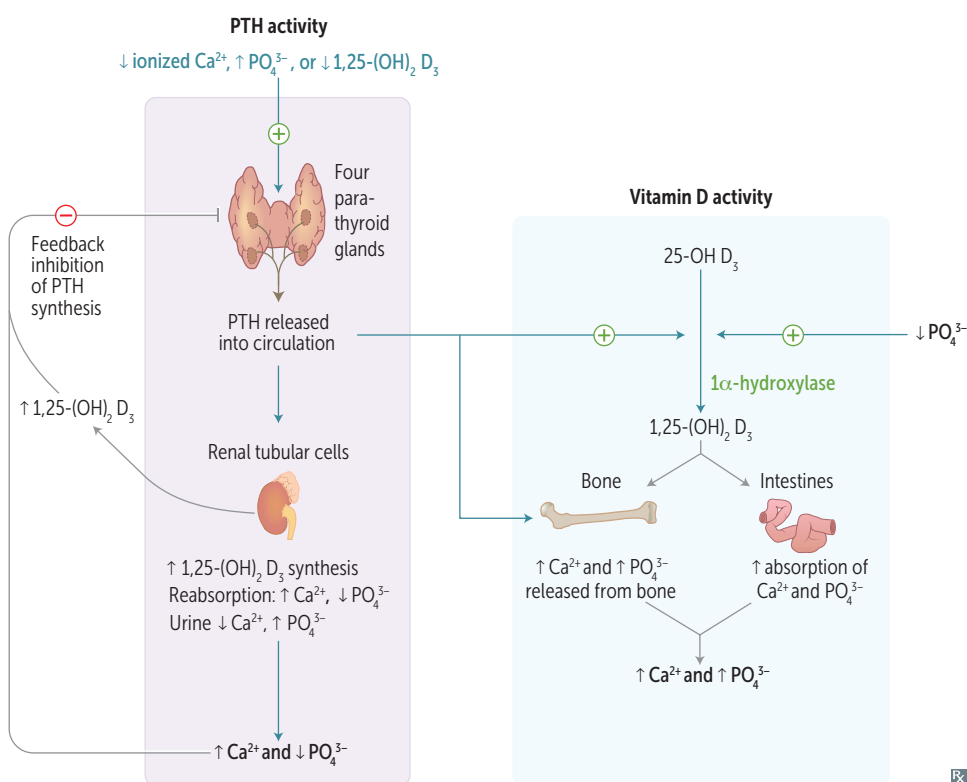
PTH = **P**hosphate-**T**rashing **H**ormone

PTH-related peptide (PTHrP) functions like PTH and is commonly increased in malignancies (eg, squamous cell carcinoma of the lung, renal cell carcinoma)

REGULATION

$\downarrow$ serum $\text{Ca}^{2+} \rightarrow \uparrow$ PTH secretion
 $\uparrow$ serum $\text{PO}_4^{3-} \rightarrow \uparrow$ PTH secretion
 $\downarrow$ serum $\text{Mg}^{2+} \rightarrow \uparrow$ PTH secretion
 $\downarrow\downarrow$ serum $\text{Mg}^{2+} \rightarrow \downarrow$ PTH secretion
 Common causes of $\downarrow$ Mg^{2+} include diarrhea, aminoglycosides, diuretics, alcohol use disorder

Ca^{2+} is the major regulator of PTH release



Calcium homeostasis

Plasma Ca^{2+} exists in three forms:

- Ionized/free ($\sim 45\%$, active form)
- Bound to albumin ($\sim 40\%$)
- Bound to anions ($\sim 15\%$)

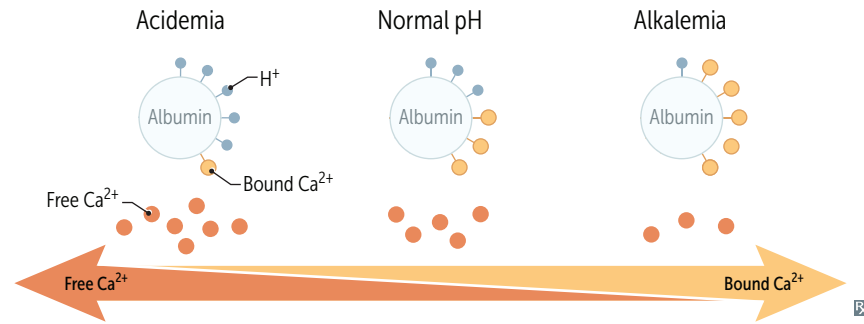
Ionized/free Ca^{2+} is 1° regulator of PTH; changes in pH alter PTH secretion, whereas changes in albumin concentration do not

Ca^{2+} competes with H^+ to bind to albumin

$\uparrow \text{pH}$ (less H^+) $\rightarrow$ albumin binds more

$\text{Ca}^{2+} \rightarrow \downarrow$ ionized Ca^{2+} (eg, cramps, pain, paresthesias, carpopedal spasm) $\rightarrow \uparrow$ PTH

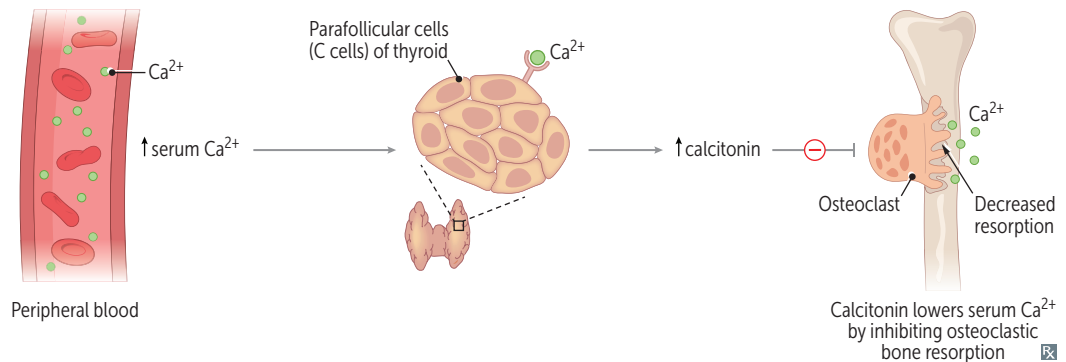
$\downarrow \text{pH}$ (more H^+) $\rightarrow$ albumin binds less Ca^{2+}
 $\rightarrow \uparrow$ ionized $\text{Ca}^{2+} \rightarrow \downarrow$ PTH



Calcitonin

SOURCE	Parafollicular cells (C cells) of thyroid.
FUNCTION	$\downarrow$ bone resorption.
REGULATION	$\uparrow$ serum $\text{Ca}^{2+} \rightarrow \uparrow$ calcitonin secretion.

Calcitonin opposes actions of PTH. Not important in normal Ca^{2+} homeostasis
 Calcitonin **tones** down serum Ca^{2+} levels and keeps it in **bones**

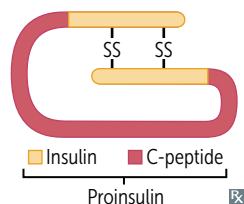


Glucagon

SOURCE	Made by α cells of pancreas.
FUNCTION	Promotes glycogenolysis, gluconeogenesis, lipolysis, ketogenesis. Elevates blood sugar levels to maintain homeostasis when bloodstream glucose levels fall too low (ie, fasting state).
REGULATION	Secreted in response to hypoglycemia. Inhibited by insulin, amylin, somatostatin, hyperglycemia.

Insulin

SYNTHESIS



Preproinsulin (synthesized in RER of pancreatic β cells) $\rightarrow$ cleavage of “presignal” $\rightarrow$ proinsulin (stored in secretory granules) $\rightarrow$ cleavage of proinsulin $\rightarrow$ exocytosis of insulin and C-peptide equally. Both insulin and C-peptide are $\uparrow$ in endogenous insulin secretion (eg, type 2 DM, insulin secretagogues, insulinoma), whereas exogenous insulin lacks C-peptide (helps differentiate factitious hypoglycemia).

Insulin is synthesized in pancreas and cleared by both liver and kidneys.

FUNCTION

Binds **insulin** receptors (tyrosine kinase activity **1**), **inducing** glucose uptake (carrier-mediated transport) **into** insulin-dependent tissue **2** and gene transcription.

Anabolic effects of insulin:

- $\uparrow$ glucose transport in skeletal muscle and adipose tissue
- $\uparrow$ glycogen synthesis and storage
- $\uparrow$ triglyceride synthesis
- $\uparrow$ Na^+ retention (kidneys)
- $\uparrow$ protein synthesis (muscles)
- $\uparrow$ cellular uptake of K^+ and amino acids
- $\downarrow$ glucagon release
- $\downarrow$ lipolysis in adipose tissue

Unlike glucose, insulin does not cross placenta. In mothers with diabetes, excess glucose can cross placenta and $\uparrow\uparrow$ fetal insulin.

Insulin-dependent glucose transporters:

- GLUT4: adipose tissue, striated muscle (exercise can also $\uparrow$ GLUT4 expression)

Insulin-independent transporters:

- GLUT1: RBCs, brain, cornea, placenta
- GLUT2 (**bidirectional**): β islet cells, liver, kidney, GI tract (think **2**-way street)
- GLUT3: brain, placenta
- GLUT5 (**fructose**): spermatocytes, GI tract
- SGLT1/SGLT2 (Na^+ -glucose cotransporters): kidney, small intestine

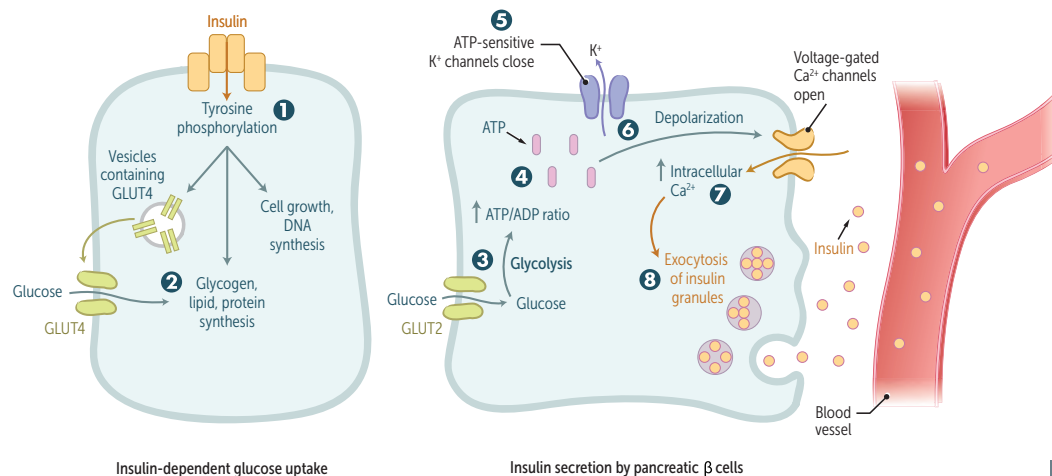
Brain prefers glucose, but may use ketone bodies during starvation. RBCs utilize only glucose, as they lack mitochondria for aerobic metabolism.

BRICK LIPS (insulin-independent glucose uptake): **B**rain, **R**BCs, **I**ntestine, **C**ornea, **K**idney, **L**iver, **I**slet (β) cells, **P**lacenta, **S**permatocytes.

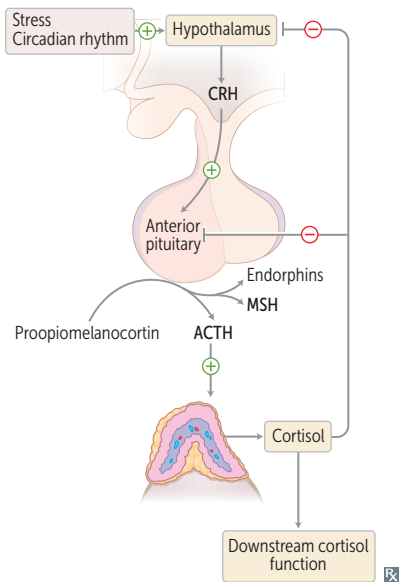
REGULATION

Glucose is the major regulator of insulin release. $\uparrow$ insulin response with oral vs IV glucose due to incretins (eg, glucagonlike peptide 1 [GLP-1], glucose-dependent insulinotropic polypeptide [GIP]), which are released after meals and $\uparrow$ β cell sensitivity to glucose. Release $\downarrow$ by α_2 , $\uparrow$ by β_2 stimulation (**2** = regulates **insulin**).

Glucose enters β cells **3** $\rightarrow$ $\uparrow$ ATP generated from glucose metabolism **4** closes K^+ channels (target of sulfonylureas) **5** and depolarizes β cell membrane **6**. Voltage-gated Ca^{2+} channels open $\rightarrow$ Ca^{2+} influx **7** and stimulation of insulin exocytosis **8**.



Cortisol

SOURCE	Adrenal zona fasciculata.	Bound to corticosteroid-binding globulin.
FUNCTION	↑ Appetite ↑ Blood pressure: - Upregulates α_1-receptors on arterioles → ↑ sensitivity to norepinephrine and epinephrine (permissive action) - At high concentrations, can bind to mineralocorticoid (aldosterone) receptors ↑ Insulin resistance (diabetogenic) ↑ Gluconeogenesis, lipolysis, and proteolysis (↓ glucose utilization) ↓ Fibroblast activity (poor wound healing, ↓ collagen synthesis, ↑ striae) ↓ Inflammatory and Immune responses: - Inhibits production of leukotrienes and prostaglandins - Inhibits WBC adhesion → neutrophilia - Blocks histamine release from mast cells - Eosinopenia, lymphopenia - Blocks IL-2 production ↓ Bone formation (↓ osteoblast activity)	Cortisol is A BIG FIB. Exogenous glucocorticoids can cause reactivation of TB and candidiasis (blocks IL-2 production). 
REGULATION	CRH (hypothalamus) stimulates ACTH release (pituitary) → cortisol production in adrenal zona fasciculata. Excess cortisol ↓ CRH, ACTH, and cortisol secretion.	Chronic stress may induce prolonged cortisol secretion, cortisol resistance, impaired immunocompetency, and dysregulation of HPA axis.

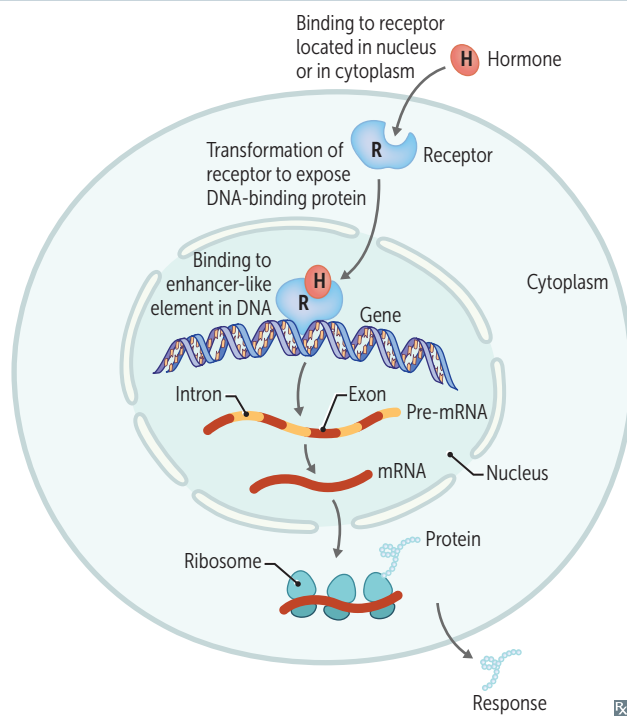
Appetite regulation

Ghrelin	Stimulates hunger (orexigenic effect) and GH release (via GH secretagog receptor). Produced by stomach. Sleep deprivation, fasting, or Prader-Willi syndrome → ↑ ghrelin production. G hrelin makes you gh row hungh ry. Acts on lateral area of hypothalamus (hunger center) to ↑ appetite.
Leptin	Satiety hormone. Produced by adipose tissue. Mutation of leptin gene → severe obesity. Obese people have ↑ leptin due to ↑ adipose tissue but are tolerant or resistant to leptin's anorexigenic effect. Sleep deprivation or starvation → ↓ leptin production. L eptin keeps you thin . Acts on ventromedial area of hypothalamus (satiety center) to ↓ appetite.
Endocannabinoids	Act at cannabinoid receptors in hypothalamus and nucleus accumbens, two key brain areas for the homeostatic and hedonic control of food intake → ↑ appetite. Exogenous cannabinoids cause “the munchies.”

Signaling pathways of endocrine hormones

cAMP	FSH, LH, ACTH, TSH, CRH, hCG, ADH (V ₂ -receptor), MSH, PTH, Calcitonin, Histamine (H ₂ -receptor), Glucagon, GHRH	FLAT ChAMPs CHuGG
cGMP	BNP, ANP, EDRF (NO)	BAD GraMP _a Think vasodilation and diuresis
IP₃	GnRH, Oxytocin, ADH (V ₁ -receptor), TRH, Histamine (H ₁ -receptor), Angiotensin II, Gastrin	GOAT HAG
Intracellular receptor	Progesterone, Estrogen, Testosterone, Cortisol, Aldosterone, T ₃ /T ₄ , Vitamin D	PET CAT in TV
Receptor tyrosine kinase	IGF-1, FGF, PDGF, EGF, Insulin	MAP kinase pathway Get Found In the MAP
Nonreceptor tyrosine kinase	G-CSF, Erythropoietin, Thrombopoietin Prolactin, Immunomodulators (eg, cytokines IL-2, IL-6, IFN), GH	JAK/STAT pathway Think acidophils and cytokines GET a JAKed PIG

Signaling pathways of steroid hormones



Steroid hormones are lipophilic and therefore must circulate bound to specific binding globulins, which ↑ their solubility.

In males, ↑ sex hormone-binding globulin (SHBG) lowers free testosterone → gynecomastia.

In females, ↓ SHBG raises free testosterone → hirsutism.

↑ estrogen (eg, OCPs, pregnancy) → ↑ SHBG.